Smallest polyomino containing all free n-ominoes (A327094)

Minimum cells in a connected polyomino on the square grid that contains every free n-omino as a subshape under rigid motion. Values proved minimal within the search window; global minimality conjectured.

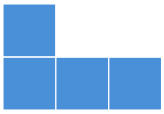
$a(1) = 1$, $a(2) = 2$, $a(3) = 4$, $a(4) = 6$, $a(5) = 9$, $a(6) = 12$, $a(7) = 17$, $a(8) = 20$, $a(9) = 26$, $a(10) = 31$

Computed by Peter Exley, May 2026

$a(1) = 1$ **[PROVED]** 1×1 bounding box, 1 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).

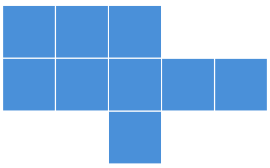
$a(2) = 2$ **[PROVED]** 1×2 bounding box, 2 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).

$a(3) = 4$ **[PROVED]** 2×3 bounding box, 4 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).

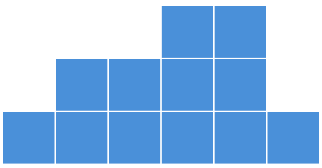


$a(4) = 6$ **[PROVED]** 2×4 bounding box, 6 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).

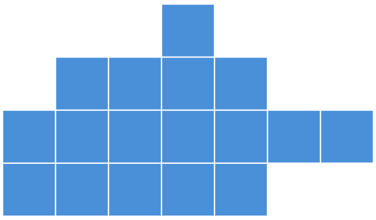
$a(5) = 9$ **[PROVED]** 3×5 bounding box, 9 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).



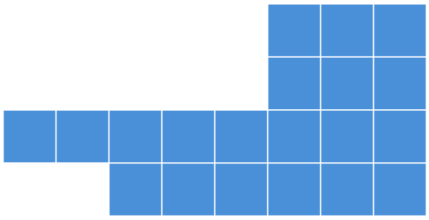
$a(6) = 12$ **[PROVED]** 3×6 bounding box, 12 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).



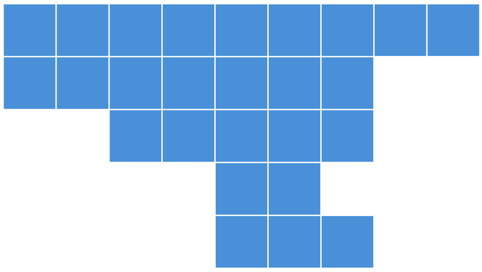
$a(7) = 17$ **[PROVED]** 4×7 bounding box, 17 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).



$a(8) = 20$ **[PROVED]** 4×8 bounding box, 20 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).



$a(9) = 26$ **[PROVED]** 5×9 bounding box, 26 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).



$a(10) = 31$ **[PROVED]** 5×10 bounding box, 31 cells. Witness found by exact container search and re-checked by an independent geometric containment verifier (all n); the lower bound is an infeasibility proof over the grid window (relaxed-LRAT refutation certificate, all n).